

Course 3 · Week 2 — Missing data, longitudinal, time series

Cheatsheet — biostats_courses

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Missingness mechanisms

Mechanism	Definition	Complete-case analysis?
MCAR	missingness independent of data	unbiased but inefficient
MAR	missingness depends on observed data	biased, imputation helps
MNAR	missingness depends on unobserved values	imputation under model only

Test is impossible without assumptions — reason about the mechanism.

Multiple imputation with **mice**

```
library(mice)
imp <- mice(df, m = 20, seed = 42, printFlag = FALSE)
fit <- with(imp, lm(y ~ x1 + x2))
pooled <- pool(fit)
summary(pooled, conf.int = TRUE)
```

Rule of thumb: $m \geq 100 \times \text{fraction of missing information}$, but 20 is a fine starting point.

Linear mixed models

```
library(lme4); library(lmerTest)
lmer(y ~ treatment * time + (1 | subject), data = df)
lmer(y ~ treatment * time + (time | subject), data = df) # random slope
```

- Random intercept: each cluster has its own baseline.
- Random slope: each cluster has its own trajectory.
- ICC $\rho = \sigma_u^2 / (\sigma_u^2 + \sigma_e^2)$.

REML for variance components; ML only for likelihood-ratio tests on fixed effects.

GLMMs & GEE

```
library(lme4)
glmer(y ~ x + (1 | cluster), data = df, family = binomial)

library(geepack)
geeglm(y ~ x, id = cluster, data = df,
       family = binomial, corstr = "exchangeable")
```

- GLMM: subject-specific effects, good for prediction.
- GEE: population-average effects, robust to working-correlation misspecification (when combined with robust SEs).

Time series basics

```
library(forecast)
ts_obj <- ts(x, frequency = 12)
decompose(ts_obj) |> plot()
auto.arima(ts_obj)

library(changepoint)
cpt.mean(x, method = "PELT") |> plot()
```

Decision rule for Week 2

- Missing data > 5% → at least sensitivity analysis; prefer MI.
- Repeated measurements → mixed model, not repeated-measures ANOVA.
- Binary outcome with clusters → GLMM (if subject effects matter) or GEE (if marginal effect is the estimand).
- Time series with trend / seasonality → decompose before modelling.

Common pitfalls

- Last-observation-carried-forward; usually biased.
- Imputing after transformation rather than in the right scale.
- Ignoring random effects when clusters are small in number (degrees of freedom correction).
- Quoting GEE coefficients as if they were subject-specific.

Further reading

- van Buuren, *Flexible Imputation of Missing Data*.
- Fitzmaurice, Laird, Ware, *Applied Longitudinal Analysis*.